Project 7000 — 1 —

Design objectives for The SIGMA Computer

A. Purpose

The primary purpose of this computer system is to fulfill our contractual obligations with the Atomic Energy Commission. A secondary, but perhaps more fundamental, purpose is to provide the world with the best general purpose computer which the latest advances in solid state components can practically produce. Care must be taken in the planning to insure consistantly high performance by all parts of the computer system. The SIGMA computer must also have a high standard for reliability and ease of maintenance in order to make its high speed effective on long and difficult problems.

B. Market

1. The SIGMA computer will be most applicable to those problems largely scientific in nature requiring a large amounts of computing time with relatively small amounts of original input and final output. There can be large amounts of high speed communication with auxiliary storage, however.

2. The immediate market will be in the Atomic Energy field in the design and simulation of nuclear weapons and reactors. A field of growing importance will be in guided missile

system design and evaluation. Later applications in the fields of chemical kinetics, economic "game theory", military operations research, real-time simulation and control of our air defense network, etc, are certain to develop when a computer capable of actually solving such problems becomes available.

3. The SIGMA arithmetic speeds and data handling capacities should be aimed at solving the problems already in existance which are not solvable on present computer systems. With advanced computer-assisted coding systems and auto-monotoring techniques, the SIGMA System should also find a market in those installations which have large quantities of medium-size problems such as the engine design simulation, flutter analysis, and control system studies in the aircraft industry. It should be noted that the existance of a computer capable of handling large problems efficiently will encourage the integration of many smaller problems done separately today into single, large problems.

~~Target Rental~~

3. The rental objective for a ~~System~~ SIGMA system consisting of a central processing unit, a console, four units of two microsecond memory (16,384 words per unit), two units of half-microsecond memory (1024 words per unit), a high speed disk file with its exchange, a regular exchange with suitable tapes and other input-output equipment should be $200,000 per month.

~~The market over a 5 year period?~~

4. There should be a market for 20 to 30 SIGMA systems over a 5 year period.

C. Fundamental Characteristics

1. The performance of the SIGMA computer on typical applications in the field of nuclear weapon or reactor design should be 100 times faster than the 704 or 709 on the internal computing portions of the problems. The input-output required for such problems must be fast enough and handled in such a way so as to not reduce this performance materially.

2. The computer will be capable of performing arithmetic in fixed point and unnormalized and normalized floating point.

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Special provisions will be made to facilitate multiple-precision arithmetic,

3, The word size shall be 64 bits and sign for data words plus the bits necessary for error detection and correction. Most instructions will be 32 bits in length, with some more complicated ones requiring 64 bits. Floating point words will contain a 48 bit mantissa plus mantissa sign, an eleven bit exponent plus exponent sign, and three tag bits,

4. The nominal arithmetic speeds should be as follows:

Fixed addition	0.2 microseconds
Floating addition	0.6
Fixed or Floating multiply	1.2
Fixed or Floating divide	1.8
Load or Store	0.2

~~The above are subject to a 25% improvement~~

The contract with the AEC allows a 25% improvement in the above times. It also states that the time required to transfer data from the memory registers to the arithmetic unit will normally add no more than 0.2 microseconds to the above times,

5, There shall be at least four units of core memory, each containing 16,384 words having a full read-write cycle time of 2.0 to 2.2 microseconds. These units will be used mainly for data storage and

will be capable of operating concurrently and asynchronously with respect to the ~~this~~ rest of the system.

6. There shall be at least two units of core memory each containing 1,024 words, with a full cycle time of 0.6 to 0.75 microseconds. These units will be used mainly for instruction storage and will also be capable of operating concurrently and asynchronously with respect to the rest of the system.

7. There shall be at least sixteen words of storage with a 0.2 to [redacted] microsecond cycle time which will be mainly used for index quantities and [redacted] storage of intermediate results.

8. To guarantee a high degree of efficiency in the arithmetic unit consecutive memory addresses will be distributed among the various storage units and a multiplexing device shall be provided to pre-reference memory so that as much of the time as is possible the data will be available without causing a delay in the arithmetic unit.

9. Automatic checking will be provided throughout the system (over). Parts of the system will be designed to detect single, double, and most multiple errors as well as to correct all single errors. During arithmetic operations the arithmetic unit will repeat its operation if an error is detected. The automatic checking, error correction, and marginal checking mechanisms will be specifically designed to reveal the nature and location of machine faults where possible.

10. An editing system will be provided to arrange input data for use by the computer and to convert computer results into the form required by output devices. The conversion of data from one radix to another or from one machine code to another must be done quickly and conveniently.

11. The computer will provide programmed access to a large number of selector devices for use in logical control. Some of these selectors will be set by the status of parts of the machine; some will be set by the data; and others can be set from the operator's console or by the program.

12. The computer will include new and powerful logical instructions including all of the sixteen logical connectives which may be applied under program control upon any addressable 64 bit word in the computer. The instruction set and formats must be chosen to insure that the SIGMA system has the greatest possible effectiveness on large and complex programs. The instruction set must also be in harmony with the requirements of machine-assisted coding.

D. Minimum Performance Criteria:

The minimum performance criteria are described in a memo dated December 30, 1957, by D.A. Jameson, E.H. Carlstrom, and H.G. Kolsky

E. Installation Considerations

1. The SIGMA System will be of such a size and layout that it can be installed in a room 54 feet long by 54 feet wide with a ceiling 8 feet high. Raised flooring should not be necessary.

2. The power and air conditioning requirements will not be more than the requirements for 3 704 systems. The solid state components should allow much lower requirements than these in practice.

3. The noise level at the operator's console will be as low as is reasonably possible.